Matching Lower Tails for Uniform Stability

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Abstract

Uniform stability now gives the logarithmic-free upper tail

$$O\Big(\gamma\log(1/\delta) + L\sqrt{\log(1/\delta)/n}\Big)$$

for every γ-stable algorithm with loss in $[0, L]$. Whether a bounded-loss learner can realize the linear confidence term has remained open. We solve this lower-bound problem. For all sufficiently large n, every $L > 0$, and every $0 \le \gamma \le L$, one finite learning problem and one deterministic uniformly stable algorithm satisfy, simultaneously for all $p \ge 2$,

$$\|R(A_S) - \widehat{R}_S(A_S)\|_p \gtrsim L \wedge \Big(\gamma p + L\sqrt{p/n}\Big).$$

The lower theorem is self-contained. Combined with the recent upper bound, it gives an exact minimax moment characterization. The construction selects the largest among exponentially many independent empirical Rademacher scores, but scales the selected coordinate by the spacing between the largest and second-largest scores. Hard selection becomes stable because a replacement can change the winner only when the two spacings have small total size. Meanwhile, an isolated upper order statistic has probability $\exp(-\Theta(p))$, spacing $\Theta(p)$, and empirical correlation bounded away from zero. This produces a bounded stable loss with gap $\Theta(\gamma p)$. An independent fixed loss supplies the sampling term on the same event. A finite-class barrier shows that exponential cardinality is necessary for range-scale deviations in the regime where our construction saturates. The result also yields matching confidence quantiles at every sufficiently small failure probability, including loss-range saturation, and closes the bounded-loss lower-bound questions posed in 2020 and 2021.

1 INTRODUCTION

Uniform stability controls how much the loss of a learned predictor can change when one training observation is replaced. It provides generalization guarantees directly from the sensitivity of the learned loss (Bousquet and Elisseeff, 2002). Its modern high-probability theory began with the observation that the classical bounded-difference analysis is much too pessimistic when the stability parameter γ is larger than order L/n (Feldman and Vondrák, 2018, 2019).

A sequence of increasingly sharp arguments has reduced the stability contribution to the generalization error. For a loss bounded by L, Bousquet et al. (2020) obtained the moment bound

$$\|R(A_S) - \widehat{R}_S(A_S)\|_p \lesssim \gamma p \log n + L\sqrt{p/n}.$$

Their proof reduced learning to a concentration problem for weakly interacting functions. They also constructed functions with moments $\Omega(\gamma p)$, but each function could be as large as $n\gamma$. It was therefore not the loss of a bounded learning problem. They explicitly asked whether the same tail can occur when the loss is uniformly bounded.

The gap persisted on both sides. Liu and Lu (2021) constructed a bounded-loss stable learner with generalization error $\Omega(\gamma + L/\sqrt{n})$ at constant probability, and left the high-confidence extension open. Very recently, Nguyen-Cung and Nguyen (2026) removed the last $\log n$ from the upper bound:

$$\|R(A_S) - \widehat{R}_S(A_S)\|_p \le 33\gamma p + L\sqrt{2p/n}. \qquad (1)$$

Their paper again separates this completed upper-bound problem from the bounded-loss learning lower bound. Our lower theorem does not use their argument. Equation (1) is needed only to turn the new lower result into a two-sided minimax characterization.

We close the lower-bound problem. Define $\mathcal{W}_p(n, \gamma, L)$ as the supremum of the L_p norm of the absolute generalization gap over all data distributions, losses in $[0, L]$, and deterministic γ-uniformly stable algorithms trained on n observations. Our main result is the following characterization.

Theorem 1 (Minimax moments)**.** *There are universal constants $c, C > 0$ and $n_0 \in \mathbb{N}$ such that, for every $n \geq n_0$, $L > 0$, $0 \leq \gamma \leq L$, and $p \geq 2$,*

$$c\left[L \wedge \left(\gamma p + L\sqrt{p/n}\right)\right] \leq \mathcal{W}_p(n, \gamma, L) \leq C\left[L \wedge \left(\gamma p + L\sqrt{p/n}\right)\right]. \tag{2}$$

The lower bounds for every p are attained by the same finite learning problem and the same deterministic algorithm.

The upper bound is (1) combined with the loss range. The contribution is the simultaneous lower bound. It gives the first bounded-loss learner that realizes the subexponential stability term. The corresponding confidence statement is also sharp throughout the full range.

Corollary 2 (Minimax confidence)**.** *For the lower-bound problem in Theorem 1, there are universal constants $c_0, c_1, C_1 > 0$ such that, for every $0 < \delta < c_0$,*

$$\mathbb{P}\left\{\operatorname{gen}(S) \geq c_1\left[L \wedge \left(\gamma \log\frac{c_0}{\delta} + L\sqrt{\frac{1}{n}\log\frac{c_0}{\delta}}\right)\right]\right\} \geq \delta. \tag{3}$$

Conversely, every bounded-loss γ-stable algorithm satisfies $|\operatorname{gen}(S)| \leq C_1[L \wedge \{\gamma \log(1/\delta) + L\sqrt{\log(1/\delta)/n}\}]$ with probability at least $1 - \delta$.

The construction is an order-statistic mechanism. It has exponentially many independent Rademacher coordinates. The algorithm chooses the coordinate with largest empirical sum and uses only the gap between the largest and second-largest sums as its amplitude. There are two complementary facts:

1. If one replacement changes the winning coordinate, the old and new winner gaps have total size at most four. Scaling the gap by $\gamma/4$ therefore makes hard selection uniformly stable.
2. At a fixed upper binomial quantile, the event that exactly one coordinate lies d lattice steps above all others has probability $\exp(-\Theta(d))$. On that event, the selected empirical correlation is constant while its stable amplitude is $\Theta(\gamma d)$.

Thus the rare quantity is a *spacing*, not the correlation itself. This distinction avoids the clipping obstruction in the natural quadratic construction. If one uses a single score X and amplitude γX, boundedness clips precisely when γp begins to dominate $L\sqrt{p/n}$. Here the selected score stays of order n, while only its spacing varies over the tail.

Table 1: Universal bounds for loss range $[0, L]$, with $p \asymp \log(1/\delta)$. Logarithmic factors in n are displayed.

work	type	stability term
Bousquet–Elisseeff (2002)	upper	$\gamma\sqrt{np}$
Feldman–Vondrák (2018)	upper	$\sqrt{\gamma L p}$
Feldman–Vondrák (2019)	upper	$\gamma \log n \log(n/\delta)$
Bousquet et al. (2020)	upper	$\gamma p \log n$
Liu–Lu (2021)	lower	γ, constant probability
Nguyen–Nguyen (2026)	upper	γp
this work	lower	γp

The result is information theoretic. The hypothesis class is finite and its log-cardinality is $\Theta(n)$. Proposition 7 shows that this order is necessary to reach a constant fraction of the loss range with the probability supplied by the stability tail whenever $n\gamma/L$ is sufficiently large. The algorithm is deterministic, and stability holds under worst-case replacement for every training sample and every test point. No average or distribution-dependent stability convention is used.

2 SETUP AND PRIOR LOWER BOUNDS

Let $S = (Z_1, \ldots, Z_n) \sim P^n$ on a measurable space $\mathcal{Z}$. A deterministic algorithm returns $A_S \in \mathcal{H}$, and a loss $\ell : \mathcal{H} \times \mathcal{Z} \to [0, L]$ defines

$$R(A_S) = \mathbb{E}_{Z \sim P}\ell(A_S, Z),$$
$$\widehat{R}_S(A_S) = \frac{1}{n}\sum_{i=1}^{n}\ell(A_S, Z_i).$$

We write $\operatorname{gen}(S) = R(A_S) - \widehat{R}_S(A_S)$. The algorithm is γ-uniformly stable if

$$\sup_{z \in \mathcal{Z}} |\ell(A_S, z) - \ell(A_{S'}, z)| \leq \gamma \tag{4}$$

whenever S and S' differ in one coordinate.

Several recent directions vary a different axis of this setup. Coordinate-dependent pointwise stability replaces γ by a root-mean-square profile and also treats gradient generalization (Fan and Lei, 2024). For randomized learners, L_2 stability and PAC-Bayes stability average over internal randomness or optimization paths (Yuan and Li, 2023; Zhou et al., 2023). Finite-moment stability permits unbounded random changes and consequently produces Gaussian-plus-polynomial upper tails (Lei et al., 2026). Other work uses additional risk geometry or optimization structure to obtain fast excess-risk bounds or to design stable ERM algorithms (Zhu et al., 2025; Vary et al., 2025). These results are complementary: none gives a lower tail for the ordinary resubstitution gap under bounded loss and the deterministic worst-case condition (4).

The closest selection mechanism is the inflated argmax of Soloff et al. (2025). It returns a set of candidate labels so nearby score vectors have overlapping outputs, using a leave-one-out average notion of selection stability. Our rule retains one winner and instead multiplies its loss contribution by the top-two gap. This enforces the pointwise loss condition (4) under every replacement and creates the required lower tail.

The generic lower construction of Bousquet et al. (2020) is useful for seeing the difficulty. For independent signs ϵ_i, their weakly interacting summands have the form

$$g_i = M\epsilon_i + \frac{\beta}{2}\epsilon_i \sum_{j \neq i} \epsilon_j.$$

Their sum contains $\frac{\beta}{2}\{(\sum_i \epsilon_i)^2 - n\}$ and hence has L_p norm of order βnp. Yet $|g_i|$ can be order βn. Replacing it by a clipped function preserves the quadratic behavior only when $\beta\sqrt{np} \lesssim L$, exactly the range in which the ordinary sampling term is at least as large as βp.

The bounded construction of Liu and Lu (2021) uses many signed coordinate vectors and predicts with fixed magnitude γ toward the sample majority. This gives a constant-probability gap of order γ. Increasing the prediction magnitude with a raw count would again encounter clipping. Our winner-spacing amplitude grows on rare events while remaining stable at every selection boundary.

3 THE WINNER-SPACING LEARNER

We first isolate the deterministic stability mechanism. For $x \in \mathbb{R}^M$, let $j(x)$ be the smallest index attaining its maximum. Let $x_{(1)} \geq x_{(2)}$ be its two largest order statistics and $D(x) = x_{(1)} - x_{(2)}$.

Lemma 3 (Stable hard selection)**.** *Suppose $M \geq 2$, $b \geq 0$, and $\|x - x'\|_\infty \leq 2$. For $a(x) = b \wedge \{\gamma D(x)/4\}$ and every $v \in \{-1, 1\}^M$,*

$$|a(x)v_{j(x)} - a(x')v_{j(x')}| \leq \gamma.$$

Proof. If $j(x) = j(x')$, each of the first two order statistics moves by at most two. Thus $|D(x) - D(x')| \leq 4$, and the claim follows because truncation is nonexpansive.

Otherwise, write $j = j(x)$ and $k = j(x')$. Since j wins before the perturbation and k wins after it,

$$D(x) \leq x_j - x_k, \qquad D(x') \leq x'_k - x'_j.$$

Adding these inequalities gives

$$D(x) + D(x') \leq (x_j - x'_j) + (x'_k - x_k) \leq 4.$$

Therefore $a(x) + a(x') \leq \gamma$, which bounds the loss difference even when v_j and v_k have opposite signs. □

We now define the learning problem. Let

$$Z = (V, U), \quad V \sim \mathrm{Unif}\{-1, 1\}^M, \quad U \sim \mathrm{Unif}\{-1, 1\}.$$

with all coordinates independent. Given S, define the coordinate scores

$$X_j = \sum_{i=1}^n V_{ij}, \qquad j \in [M].$$

The algorithm outputs $A_S = (j_*, a)$, where

$$j_* = j(X), \qquad a = \frac{L}{4} \wedge \frac{\gamma}{4}\{X_{(1)} - X_{(2)}\}. \tag{5}$$

Only $n + 1$ amplitudes can occur, so the hypothesis class can be taken as the finite set of attainable pairs (j_*, a), with cardinality at most $M(n + 1)$. The loss is

$$\ell((j, a), (v, u)) = \frac{L}{4} - av_j + \frac{L}{4} - \frac{L}{4}u. \tag{6}$$

Both components lie in $[0, L/2]$, so $0 \leq \ell \leq L$. Replacing one observation changes every X_j by at most two. Lemma 3 therefore proves γ-uniform stability.

The population risk is always $L/2$. The empirical score of the selected coordinate and the independent sampling sign give the exact identity

$$\mathrm{gen}(S) = \frac{aX_{(1)}}{n} + \frac{L}{4n}\sum_{i=1}^n U_i. \tag{7}$$

4 AN EXPONENTIAL SPACING EVENT

Let $K_j = (X_j + n)/2$. The K_j are independent $\mathrm{Bin}(n, 1/2)$ variables. Set

$$k_0 = \left\lceil \frac{3n}{4} \right\rceil, \quad b = \mathbb{P}\{K_1 \geq k_0 + 1\}, \quad M = \lfloor b^{-1} \rfloor. \tag{8}$$

Thus $\log M = \Theta(n)$. For $d \geq 1$, consider the event E_d that exactly one coordinate has $K_j \geq k_0 + d$ and every other coordinate has $K_j \leq k_0$.

Lemma 4 (Binomial spacing)**.** *If $n \geq 64$ and $1 \leq d \leq n/16$, then*

$$\mathbb{P}(E_d) \geq \frac{5}{12}5^{-d}.$$

On E_d, $X_{(1)} \geq n/2$ and $X_{(1)} - X_{(2)} \geq 2d$.

The second claim is immediate. The probability calculation is short enough to include. Let $a_d = \mathbb{P}\{K_1 \geq k_0 + d\}$. Independence gives

$$\mathbb{P}(E_d) = M a_d (1-b)^{M-1}.$$

Here $M \geq 1/(2b)$ and $(1-b)^{M-1} \geq 1/4$. Above k_0, successive binomial masses decrease by at most a factor five for $d \leq n/16$, while the tail b is at most $3/2$ times its first mass. Hence

$$\frac{a_d}{b} \geq \frac{2}{3} 5^{-(d-1)},$$

which proves the lemma. Supplementary Section A.1 verifies the two numerical inequalities without suppressed endpoint terms.

On E_d, (5) and (7) imply

$$\frac{aX_{(1)}}{n} \geq \min\left\{\frac{L}{8}, \frac{\gamma d}{4}\right\}. \tag{9}$$

This is the missing stability tail. The winning score is already order n. Every additional isolated lattice step increases the loss amplitude by order γ, and d isolated steps cost only exponential probability.

For the other term, a standard binomial lower estimate gives the following form. A self-contained proof is in Supplementary Section A.2.

Lemma 5 (Sampling anti-concentration)**.** *There are universal $c_s, C_s > 0$ such that, for $n \geq 64$ and $1 \leq d \leq n/64$,*

$$\mathbb{P}\left\{\sum_{i=1}^{n} U_i \geq \sqrt{nd}\right\} \geq c_s e^{-C_s d}.$$

The events in Lemmas 4 and 5 are independent. Combining them with (7) proves the key simultaneous tail statement.

Theorem 6 (One construction, all scales)**.** *There are universal $c, C > 0$ such that, for the learner (5)–(6), every integer $1 \leq d \leq n/64$ satisfies*

$$\mathbb{P}\left\{ \operatorname{gen}(S) \geq \min\left(\frac{L}{8}, \frac{\gamma d}{4}\right) + \frac{L}{4}\sqrt{\frac{d}{n}} \right\} \geq c e^{-Cd}. \tag{10}$$

The exponential coordinate family is forced by the range-scale part of the claim, rather than by the proof technique. The following elementary barrier makes this precise.

Proposition 7 (Complexity needed for saturation)**.** *Let $\mathcal{H}$ be finite with $N = |\mathcal{H}|$, let every $\ell(h, \cdot)$ take values in $[0, L]$, and let $A_S \in \mathcal{H}$ be any data-dependent output. For every $u \geq 0$,*

$$\mathbb{P}\left\{ |\operatorname{gen}(S)| \geq L\sqrt{\frac{\log(2N) + u}{2n}} \right\} \leq e^{-u}. \tag{11}$$

For our learner, if $\gamma > 0$ and $d = \lceil L/(2\gamma) \rceil \leq n/16$, then

$$\mathbb{P}\{\operatorname{gen}(S) \geq L/8\} \geq \frac{5}{24} 5^{-d}. \tag{12}$$

Consequently, any finite-class learner with the probability in (12) *must have*

$$\log N \geq \frac{n}{32} - d\log 5 + \log(5/48).$$

In particular, exponential cardinality is necessary when $L/\gamma = o(n)$.

Proof. For each fixed h, Hoeffding's inequality bounds the two-sided deviation by $2\exp(-2nt^2/L^2)$. A union bound over $\mathcal{H}$ proves (11), even though A_S depends on S. For the second claim, Lemma 4 and the choice of d make the first term in (7) at least $L/8$. Independently, $\sum_i U_i \geq 0$ with probability at least $1/2$. This proves (12). Comparing it with the fixed-class union bound at $t = L/8$ gives the displayed cardinality inequality. ☐

5 MINIMAX CONSEQUENCES

We prove Theorem 1. The upper bound follows from (1) and $|\operatorname{gen}(S)| \leq L$.

For the lower bound, first suppose $2 \leq p \leq n/64$ and take $d = \lfloor p \rfloor$. Theorem 6 and $\|Y\|_p \geq t\mathbb{P}\{|Y| \geq t\}^{1/p}$ give

$$\|\operatorname{gen}(S)\|_p \geq c\left[\min(L, \gamma p) + L\sqrt{p/n}\right].$$

Since the sampling term is at most $L/8$ in this range, the right side is at least a universal multiple of $L \wedge (\gamma p + L\sqrt{p/n})$.

If $p > n/64$, use Theorem 6 at an integer d of order n. Its sampling component is a constant multiple of L, its probability is at least e^{-Cn}, and taking the pth root loses only a universal factor. Monotonicity of L_p norms gives the same conclusion. This proves (2). Notice that the dimension M and the algorithm do not depend on p.

For Corollary 2, choose d proportional to $\log(c_0/\delta)$ while this is below $n/64$, and use (10). At smaller δ, keep d of order n. The sampling component has already reached a constant fraction of L, which is the range cap. The reverse inequality is the high-probability consequence of (1). Full constant bookkeeping appears in Supplementary Section A.4.

6 DISCUSSION

The construction clarifies why boundedness does not improve the universal stability tail. A clipped quadratic learner ties the amplitude and empirical correlation to the same score. Clipping then intervenes at the sampling crossover. Winner-spacing separates these roles. Extreme-value selection keeps one coordinate macroscopically correlated with the sample. The top-two spacing remains a locally Lipschitz amplitude with an exponential tail.

The proof also identifies a reusable stability primitive. Hard selection among data-dependent candidates is normally discontinuous. Multiplying the winner by its certificate of uniqueness makes the output Lipschitz, because a change of winner certifies that both margins were small. This principle may be useful for stable model selection beyond the lower-bound setting.

Our result is a worst-case characterization of what uniform stability alone implies. Additional structure can yield stronger guarantees. Examples include Bernstein conditions for excess risk (Klochkov and Zhivotovskiy, 2021), locally elastic stability (Deng et al., 2021), and algorithm-specific properties of regularized, iterative, or fixed-precision methods (Vary et al., 2025; Shin and Park, 2026). The theorem shows that any universal improvement must use such information. Uniform loss boundedness by itself is not enough.

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A COMPLETE PROOFS

A.1 Endpoint details for the spacing lemma

We prove Lemma 4. Write

$$q_k = \mathbb{P}\{\mathrm{Bin}(n, 1/2) = k\} = 2^{-n}\binom{n}{k}.$$

For $k \geq k_0 + 1$,

$$\frac{q_{k+1}}{q_k} = \frac{n-k}{k+1} \leq \frac{1}{3}.$$

Consequently,

$$b = \sum_{k=k_0+1}^{n} q_k \leq \frac{3}{2} q_{k_0+1}. \tag{A.1}$$

If $1 \leq r \leq d-1$ and $d \leq n/16$, then $n \geq 64$ gives

$$\begin{aligned} \frac{q_{k_0+r+1}}{q_{k_0+r}} &= \frac{n-k_0-r}{k_0+r+1} \\ &\geq \frac{3n/16}{13n/16+1} \geq \frac{1}{5}. \end{aligned}$$

It follows that

$$a_d \geq q_{k_0+d} \geq q_{k_0+1} 5^{-(d-1)} \geq \frac{2}{3} b 5^{-(d-1)}. \tag{A.2}$$

The number b is at most $1/2$. Since $M = \lfloor 1/b \rfloor$, we have $M \geq 1/(2b)$. Also $M - 1 \leq 1/b$, so

$$(1-b)^{M-1} \geq (1-b)^{1/b} \geq \frac{1}{4},$$

where the last inequality holds for $0 < b \leq 1/2$. Summing over the unique exceptional coordinate and using independence,

$$\begin{aligned} \mathbb{P}(E_d) &= M a_d (1-b)^{M-1} \\ &\geq \frac{a_d}{8b} \geq \frac{5}{12} 5^{-d}. \end{aligned}$$

This proves the probability claim. If E_d occurs, then the winning binomial count is at least $k_0 + d$, every other count is at most k_0, and hence

$$X_{(1)} \geq 2k_0 - n \geq n/2, \qquad X_{(1)} - X_{(2)} \geq 2d.$$

For completeness, Hoeffding's inequality gives $b \leq e^{-n/8}$, while $b \geq q_{k_0+1} \geq 2^{-n}$. Together with $1/(2b) \leq M \leq 1/b$, these inequalities give $e^{cn} \leq M \leq e^{Cn}$ for universal positive constants and all sufficiently large n. Thus the hypothesis class is finite and $\log M = \Theta(n)$.

A.2 A binomial interval lower bound

We prove Lemma 5. We use the following elementary estimate.

Lemma A.1. *There are universal constants $c, C > 0$ such that, for $T \sim \mathrm{Bin}(n, 1/2)$ and every integer r with $|r| \leq n/8$,*

$$\mathbb{P}\{T = \lfloor n/2 \rfloor + r\} \geq \frac{c}{\sqrt{n}} \exp\left(-C\frac{r^2}{n}\right).$$

Proof. Write $m = \lfloor n/2 \rfloor$. The elementary central binomial estimate gives

$$2^{-n}\binom{n}{m} \geq (2n)^{-1/2}.$$

For even $n = 2m$ and $r > 0$, comparison with the central mass gives the exact product

$$\frac{q_{m+r}}{q_m} = \prod_{s=1}^{r} \frac{m-s+1}{m+s} = \prod_{s=1}^{r} \left(1 - \frac{2s-1}{m+s}\right).$$

When $r \le n/8$, every fraction subtracted from one is at most $2/5$. Since $\log(1-x) \ge -x/(1-x)$ for $0 \le x < 1$,

$$\log \frac{q_{m+r}}{q_m} \ge -\frac{5}{3} \sum_{s=1}^{r} \frac{2s-1}{m+s} \ge -\frac{10r^2}{3n}.$$

For odd $n = 2m+1$, the two central masses are equal and, for $r > 0$,

$$\frac{q_{m+r}}{q_m} = \prod_{s=1}^{r} \frac{m-s+2}{m+s} = \prod_{s=1}^{r} \left(1 - \frac{2s-2}{m+s}\right).$$

The first factor is one and the same calculation applies. By symmetry, $q_{m-r} = q_{m+1+r}$, which introduces at most one extra factor bounded below by e^{-1} in the stated range. Thus the lemma holds, for example, with $c = 2^{-1/2}e^{-1}$ and $C = 8$. □

Let $t = \sqrt{nd}$. Consider the parity-compatible values of $\sum_i U_i$ in $[t, t+\sqrt{n}/2]$. There are at least $c\sqrt{n}$ such values. If $d \le n/64$ and $n \ge 64$, then every corresponding binomial displacement from $n/2$ has magnitude at most $n/8$. Lemma A.1 bounds each mass below by

$$\frac{c}{\sqrt{n}} \exp\{-C(d+1)\}.$$

Summing the masses proves

$$\mathbb{P}\left\{\sum_i U_i \ge \sqrt{nd}\right\} \ge c_s e^{-C_s d}.$$

A.3 Uniform stability in all tie cases

We expand the proof of Lemma 3. The first and second order-statistic maps are each one-Lipschitz in the sup norm. Therefore D is two-Lipschitz. Under $\|x - x'\|_\infty \le 2$, this gives $|D(x) - D(x')| \le 4$.

If the tie-broken maximizer does not change, the selected signed coordinate is the same and

$$|a(x)v_{j(x)} - a(x')v_{j(x')}| = |a(x) - a(x')| \le \gamma.$$

This includes cases where the identity of the runner-up changes.

If the maximizer changes, set $j = j(x)$ and $k = j(x')$. The definition of second largest, including ties, gives

$$D(x) \le x_j - x_k, \qquad D(x') \le x'_k - x'_j.$$

The right sides are nonnegative because j and k win in their respective vectors. Hence

$$\begin{aligned} D(x) + D(x') &\le (x_j - x_k) + (x'_k - x'_j) \\ &= (x_j - x'_j) + (x'_k - x_k) \le 4. \end{aligned}$$

Truncation can only decrease the amplitudes, so

$$\begin{aligned} |a(x)v_j - a(x')v_k| &\le a(x) + a(x') \\ &\le \frac{\gamma}{4}\{D(x) + D(x')\} \le \gamma. \end{aligned}$$

The argument covers a transition into or out of any tie because its corresponding gap is zero.

A.4 Moment and confidence conversions

Let $G = \mathrm{gen}(S)$ for the construction in Section 3. Theorem 6 supplies constants $c_0, C_0 > 0$ such that

$$\mathbb{P}\left\{G \geq \min\left(\frac{L}{8}, \frac{\gamma d}{4}\right) + \frac{L}{4}\sqrt{\frac{d}{n}}\right\} \geq c_0 e^{-C_0 d} \tag{A.3}$$

for $1 \leq d \leq n/64$.

Suppose first that $2 \leq p \leq n/64$ and set $d = \lfloor p \rfloor$. Then $d \geq p/2$, and

$$(c_0 e^{-C_0 d})^{1/p} \geq c_0^{1/2} e^{-C_0}.$$

Therefore

$$\|G\|_p \geq c\left\{\min(L, \gamma p) + L\sqrt{p/n}\right\}. \tag{A.4}$$

The term $L\sqrt{p/n}$ is at most $L/8$. A two-case check according to whether $\gamma p \geq L$ now gives

$$\|G\|_p \geq c\left[L \wedge \left(\gamma p + L\sqrt{p/n}\right)\right].$$

For $p > n/64$, choose an integer $d_* = \lfloor n/64 \rfloor$, increasing n_0 if needed. The sampling term in (A.3) is at least cL, and

$$(c_0 e^{-C_0 d_*})^{1/p} \geq c$$

because $d_*/p \leq 1$. Thus $\|G\|_p \geq cL$. At the same time, $L\sqrt{p/n} \geq cL$, so this is the required range-saturated lower bound.

For confidence, fix a sufficiently small $c_1 > 0$ and set

$$d = 1 \vee \min\left\{\left\lfloor c_1 \log \frac{c_0}{\delta} \right\rfloor, \left\lfloor \frac{n}{64} \right\rfloor\right\}.$$

Choosing c_1 from C_0 makes the probability in (A.3) at least δ. If the logarithm is below a constant multiple of n, the threshold in (A.3) is a universal multiple of the right side in (3). If it is larger, the sampling term at d is a constant multiple of L, which equals the range-truncated target. Adjusting constants handles the bounded interval of δ for which $d = 1$.

For the converse, apply (1) to $p = 2 \vee \log(1/\delta)$. Markov's inequality in the form

$$\mathbb{P}\{|G| \geq e\|G\|_p\} \leq e^{-p}$$

gives the asserted upper confidence bound, and $|G| \leq L$ supplies its range truncation.

B ADDITIONAL COMPARISONS

The lower bound concerns the ordinary resubstitution generalization gap of one deterministic learner. This distinguishes it from three nearby statements.

First, Proposition 9 of Bousquet et al. (2020) proves sharpness of a product-space concentration theorem. Their summands obey the required cross-coordinate bounded differences, but their absolute magnitudes scale as $n\gamma$. Lemma 7 in that paper maps a bounded stable learner to such summands, not conversely. Our construction supplies the missing reverse example at the level of a learning problem.

Second, Liu and Lu (2021) prove the bounded-loss lower bound only at constant probability. Their predictor uses fixed magnitude γ along each observed coordinate. The present amplitude is data dependent but remains stable because it vanishes at selection ties. Theorem 6 recovers the constant-probability endpoint and all smaller failure probabilities with one learner.

Third, the lower bound does not contradict stronger algorithm-specific results. A stable algorithm may possess convexity, a Bernstein condition, a small hypothesis class, or distribution-dependent sensitivity beyond (4). The supremum in $\mathcal{W}_p$ deliberately removes such assumptions, exactly as the universal upper bounds do.